Practica 2.0 Despliegue Apache

Revisar la configuración del sitio

```
david@Despliegue:~$ sudo cat /etc/apache2/sites-enabled/000-default.conf
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerAdmin webmaster@localhost
        DocumentRoot /var/www/html

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
david@Despliegue:~$
```

Habilitar módulos de apache

```
david@Despliegue:~$ sudo a2enmod proxy_fcgi setenvif
Considering dependency proxy for proxy_fcgi:
Enabling module proxy.
Enabling module proxy_fcgi.
Module setenvif already enabled
To activate the new configuration, you need to run:
  systemctl restart apache2
david@Despliegue:~$
```

```
david@Despliegue:~$ sudo a2enconf php8.5-fpm
Enabling conf php8.5-fpm.
To activate the new configuration, you need to run:
  systemctl reload apache2
david@Despliegue:~$
```

```
david@Despliegue:~$ sudo systemctl reload apache2
david@Despliegue:~$
```

Verificar que los módulos están habilitados

```
david@Despliegue:~$ sudo apache2ctl -M
[sudo: authenticate] Password:
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'Serve
rName' directive globally to suppress this message
Loaded Modules:
 core_module (static)
 so_module (static)
 watchdog_module (static)
 http_module (static)
 log_config_module (static)
 logio_module (static)
 version_module (static)
 unixd_module (static)
 systemd_module (static)
 access_compat_module (shared)
 alias_module (shared)
 auth_basic_module (shared)
 authn_core_module (shared)
 authn_file_module (shared)
 authz_core_module (shared)
 authz_host_module (shared)
 authz_user_module (shared)
 autoindex_module (shared)
 deflate_module (shared)
 dir_module (shared)
 env_module (shared)
 filter_module (shared)
 mime_module (shared)
 mpm_event_module (shared)
 negotiation_module (shared)
 proxy_module (shared)
 proxy_fcgi_module (shared)
 reqtimeout_module (shared)
 setenvif_module (shared)
 status_module (shared)
```

Comprobar la configuración PHP-FPM

```
david@Despliegue:~$ ls -l /etc/apache2/conf-enabled/
total 0
lrwxrwxrwx 1 root root 30 Oct  1 16:56 charset.conf -> ../conf-available/charset.conf
lrwxrwxrwx 1 root root 44 Oct  1 16:56 localized-error-pages.conf -> ../conf-available/localized-error-pages.conf
lrwxrwxrwx 1 root root 46 Oct  1 16:56 other-vhosts-access-log.conf -> ../conf-available/other-vhosts-access-log.conf
lrwxrwxrwx 1 root root 33 Nov  5 19:52 php8.5-fpm.conf -> ../conf-available/php8.5-fpm.conf
lrwxrwxrwx 1 root root 31 Oct  1 16:56 security.conf -> ../conf-available/security.conf
lrwxrwxrwx 1 root root 36 Oct  1 16:56 serve-cgi-bin.conf -> ../conf-available/serve-cgi-bin.conf
david@Despliegue:~$
```

Copiar la aplicación PHP al directorio de Apache.

```
david@Despliegue:~$ sudo mkdir -p /var/www/html/running-zones
david@Despliegue:~$
```

Comprobar que aparece el directorio running-zones

```
david@Despliegue:~$ ls -la /var/www/html/
total 24
drwxr-xr-x 4 root  root  4096 Nov  5 19:53 .
drwxr-xr-x 6 root  root  4096 Nov  5 18:26 ..
drwxr-xr-x 5 root  root  4096 Oct 15 17:46 aulaweb
-rw-r--r-- 1 root  root   229 Oct  8 17:27 index.html
-rw-r--r-- 1 root  root   615 Nov  5 16:07 index.nginx-debian.html
drwxr-xr-x 4 david david 4096 Nov  5 19:35 running-zones
david@Despliegue:~$
```

Copiar la aplicación PHP a Apache

```
david@Despliegue:~$ sudo rsync -av /var/www/running-zones/ /var/www/html/running-zones/
sending incremental file list
./
02calculadoraBootstrap.php
04ArrayBootstrap.php
Instalación y configuración del servidor web NGINX.odt
Instalación y configuración del servidor web NGINX.pdf
Práctica 2.1 Despliegue de NGINX  PHP FPM.odt
mi-app-php/
mi-app-php/.env
mi-app-php/config/
mi-app-php/config/database.php
mi-app-php/public/
mi-app-php/public/index.php
running_zone/

sent 974,222 bytes  received 191 bytes  1,948,826.00 bytes/sec
total size is 973,078  speedup is 1.00
david@Despliegue:~$
```

Comprobar que los archivos están en el directorio de Apache

```
david@Despliegue:~$ ls -la /var/www/html/running-zones/
total 976
drwxr-xr-x 4 david david   4096 Nov  5 19:35  .
drwxr-xr-x 4 root  root    4096 Nov  5 19:53  ..
-rw-r--r-- 1 david david   4186 Nov  5 19:14  02calculadoraBootstrap.php
-rw-r--r-- 1 david david   1147 Nov  5 19:35  04ArrayBootstrap.php
-rw-r--r-- 1 david david 491279 Nov  5 18:20 'Instalaci'$'\303\263''n y configuraci'$'\303\263''n del servidor web NGINX
.odt'
-rw-r--r-- 1 david david 311224 Nov  5 18:20 'Instalaci'$'\303\263''n y configuraci'$'\303\263''n del servidor web NGINX
.pdf'
-rw-r--r-- 1 david david 165205 Nov  5 18:20 'Pr'$'\303\241''ctica 2.1 Despliegue de NGINX  PHP FPM.odt'
drwx------ 4 david david   4096 Nov  5 18:49  mi-app-php
drwx------ 2 david david   4096 Nov  5 18:37  running_zone
```

Configurar los permisos de la aplicación

```
david@Despliegue:~$ sudo chown -R www-data:www-data /var/www/html/running-zones
david@Despliegue:~$ sudo chmod -R 755 /var/www/html/running-zones
david@Despliegue:~$ ls -la /var/www/html/running-zones/
total 976
drwxr-xr-x 4 www-data www-data   4096 Nov  5 19:35  .
drwxr-xr-x 4 root     root       4096 Nov  5 19:53  ..
-rwxr-xr-x 1 www-data www-data   4186 Nov  5 19:14  02calculadoraBootstrap.php
-rwxr-xr-x 1 www-data www-data   1147 Nov  5 19:35  04ArrayBootstrap.php
-rwxr-xr-x 1 www-data www-data 491279 Nov  5 18:20 'Instalaci'$'\303\263''n y configuraci'$'\303\263''n del servidor web
 NGINX.odt'
-rwxr-xr-x 1 www-data www-data 311224 Nov  5 18:20 'Instalaci'$'\303\263''n y configuraci'$'\303\263''n del servidor web
 NGINX.pdf'
-rwxr-xr-x 1 www-data www-data 165205 Nov  5 18:20 'Pr'$'\303\241''ctica 2.1 Despliegue de NGINX  PHP FPM.odt'
drwxr-xr-x 4 www-data www-data   4096 Nov  5 18:49  mi-app-php
drwxr-xr-x 2 www-data www-data   4096 Nov  5 18:37  running_zone
david@Despliegue:~$
```

Comprobar la configuración de PHP-FPM en Apache

```
david@Despliegue:~$ sudo systemctl status php8.5-fpm
● php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager
     Loaded: loaded (/usr/lib/systemd/system/php8.5-fpm.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-11-12 17:10:48 UTC; 37min ago
 Invocation: a1a1120f6dc1421591e82b0e33986fca
       Docs: man:php-fpm8.5(8)
    Process: 2105 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.5/fpm/pool.>
   Main PID: 2102 (php-fpm8.5)
     Status: "Processes active: 0, idle: 2, Requests: 3, slow: 0, Traffic: 0.00req/sec"
      Tasks: 3 (limit: 7397)
     Memory: 11.2M (peak: 11.8M)
        CPU: 134ms
     CGroup: /system.slice/php8.5-fpm.service
             ├─2102 "php-fpm: master process (/etc/php/8.5/fpm/php-fpm.conf)"
             ├─2103 "php-fpm: pool www"
             └─2104 "php-fpm: pool www"

Nov 12 17:10:48 Despliegue systemd[1]: Starting php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager...
Nov 12 17:10:48 Despliegue systemd[1]: Started php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager.
lines 1-18/18 (END)
```

Después comprobar que Apache tiene habilitada la configuración

```
david@Despliegue:~$ sudo a2query -c php8.5-fpm
php8.5-fpm (enabled by site administrator)
david@Despliegue:~$
```

Y que los módulos necesarios están activos

```
david@Despliegue:~$ sudo a2query -m proxy_fcgi
proxy_fcgi (enabled by site administrator)
david@Despliegue:~$ sudo a2query -m setenvif
setenvif (enabled by maintainer script)
david@Despliegue:~$
```

Comprobar la configuración de Apache

```
david@Despliegue:~$ sudo apache2ctl configtest
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'Serve
rName' directive globally to suppress this message
Syntax OK
david@Despliegue:~$
```

Y comprobar los sitios virtuales

```
david@Despliegue:~$ sudo apache2ctl -S
AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'Serve
rName' directive globally to suppress this message
VirtualHost configuration:
*:80                   is a NameVirtualHost
         default server 127.0.1.1 (/etc/apache2/sites-enabled/000-default.conf:1)
         port 80 namevhost 127.0.1.1 (/etc/apache2/sites-enabled/000-default.conf:1)
         port 80 namevhost mipagina.local (/etc/apache2/sites-enabled/mipagina.conf:1)
ServerRoot: "/etc/apache2"
Main DocumentRoot: "/var/www/html"
Main ErrorLog: "/var/log/apache2/error.log"
Mutex watchdog-callback: using_defaults
Mutex proxy: using_defaults
Mutex default: dir="/run/apache2/" mechanism=default
PidFile: "/run/apache2/apache2.pid"
Define: DUMP_VHOSTS
Define: DUMP_RUN_CFG
User: name="www-data" id=33
Group: name="www-data" id=33
david@Despliegue:~$
```

Comprobar el estado de Apache

```
david@Despliegue:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-11-12 16:56:59 UTC; 56min ago
 Invocation: 49daa0b486c248e38dc1c2cbf80ab8a0
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 3441 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
   Main PID: 1247 (apache2)
     Status: "Total requests: 2; Idle/Busy workers 100/0;Requests/sec: 0.000588; Bytes served/sec:   0 B/sec"
      Tasks: 55 (limit: 7397)
     Memory: 8.3M (peak: 11.2M)
        CPU: 512ms
     CGroup: /system.slice/apache2.service
             ├─1247 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─3445 /usr/sbin/apache2 -k start -DFOREGROUND
             └─3446 /usr/sbin/apache2 -k start -DFOREGROUND

Nov 12 16:56:58 Despliegue systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 12 16:56:59 Despliegue apachectl[1247]: AH00558: apache2: Could not reliably determine the server's fully qualified>
Nov 12 16:56:59 Despliegue systemd[1]: Started apache2.service - The Apache HTTP Server.
Nov 12 17:35:53 Despliegue systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Nov 12 17:35:53 Despliegue apachectl[3188]: AH00558: apache2: Could not reliably determine the server's fully qualified>
Nov 12 17:35:53 Despliegue systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
Nov 12 17:50:30 Despliegue systemd[1]: Reloading apache2.service - The Apache HTTP Server...
Nov 12 17:50:30 Despliegue apachectl[3441]: AH00558: apache2: Could not reliably determine the server's fully qualified>
Nov 12 17:50:30 Despliegue systemd[1]: Reloaded apache2.service - The Apache HTTP Server.
lines 1-25/25 (END)
```

Comprobar también PHP-FPM

```
david@Despliegue:~$ sudo systemctl status php8.5-fpm
● php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager
     Loaded: loaded (/usr/lib/systemd/system/php8.5-fpm.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-11-12 17:10:48 UTC; 43min ago
 Invocation: a1a1120f6dc1421591e82b0e33986fca
       Docs: man:php-fpm8.5(8)
    Process: 2105 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.5/fpm/pool.>
   Main PID: 2102 (php-fpm8.5)
     Status: "Processes active: 0, idle: 2, Requests: 3, slow: 0, Traffic: 0.00req/sec"
      Tasks: 3 (limit: 7397)
     Memory: 10.7M (peak: 11.8M)
        CPU: 163ms
     CGroup: /system.slice/php8.5-fpm.service
             ├─2102 "php-fpm: master process (/etc/php/8.5/fpm/php-fpm.conf)"
             ├─2103 "php-fpm: pool www"
             └─2104 "php-fpm: pool www"

Nov 12 17:10:48 Despliegue systemd[1]: Starting php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager...
Nov 12 17:10:48 Despliegue systemd[1]: Started php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager.
lines 1-18/18 (END)
```

Comprobar el firewall

```
david@Despliegue:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
80/tcp                     ALLOW       Anywhere
22/tcp                     ALLOW       Anywhere
8080/tcp                   ALLOW       Anywhere
80/tcp (v6)                ALLOW       Anywhere (v6)
22/tcp (v6)                ALLOW       Anywhere (v6)
8080/tcp (v6)              ALLOW       Anywhere (v6)

david@Despliegue:~$
```

```
david@Despliegue:~$ sudo ufw allow 'Apache'
Rule added
Rule added (v6)
david@Despliegue:~$
```

Reiniciar Apache y PHP-FPM

```
david@Despliegue:~$ sudo systemctl restart apache2
david@Despliegue:~$ sudo systemctl restart php8.5-fpm
```

Comprobar ambos servicios

```
david@Despliegue:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-11-12 17:56:19 UTC; 41s ago
 Invocation: 5189dfc92c564cddba9c308d2fe72f01
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3609 (apache2)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 55 (limit: 7397)
     Memory: 5.9M (peak: 6.2M)
        CPU: 42ms
     CGroup: /system.slice/apache2.service
             ├─3609 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─3614 /usr/sbin/apache2 -k start -DFOREGROUND
             └─3615 /usr/sbin/apache2 -k start -DFOREGROUND

Nov 12 17:56:19 Despliegue systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 12 17:56:19 Despliegue apachectl[3609]: AH00558: apache2: Could not reliably determine the server's fully qualified>
Nov 12 17:56:19 Despliegue systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-18/18 (END)
```

```
david@Despliegue:~$ sudo systemctl status php8.5-fpm
● php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager
     Loaded: loaded (/usr/lib/systemd/system/php8.5-fpm.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-11-12 17:56:25 UTC; 49s ago
 Invocation: 780d535b55594295af00243599cb1afc
       Docs: man:php-fpm8.5(8)
    Process: 3686 ExecStartPost=/usr/lib/php/php-fpm-socket-helper install /run/php/php-fpm.sock /etc/php/8.5/fpm/pool.>
   Main PID: 3682 (php-fpm8.5)
     Status: "Processes active: 0, idle: 2, Requests: 0, slow: 0, Traffic: 0.00req/sec"
      Tasks: 3 (limit: 7397)
     Memory: 10M (peak: 11.7M)
        CPU: 42ms
     CGroup: /system.slice/php8.5-fpm.service
             ├─3682 "php-fpm: master process (/etc/php/8.5/fpm/php-fpm.conf)"
             ├─3684 "php-fpm: pool www"
             └─3685 "php-fpm: pool www"

Nov 12 17:56:25 Despliegue systemd[1]: Starting php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager...
Nov 12 17:56:25 Despliegue systemd[1]: Started php8.5-fpm.service - The PHP 8.5 FastCGI Process Manager.
lines 1-18/18 (END)
```

Comprobar Apache mediante curl

```
david@Despliegue:~$ curl -I http://localhost/
HTTP/1.1 200 OK
Date: Thu, 12 Nov 2026 17:57:42 GMT
Server: Apache/2.4.66 (Ubuntu)
Last-Modified: Thu, 08 Oct 2026 17:27:07 GMT
ETag: "e5-65d578a80aee0"
Accept-Ranges: bytes
Content-Length: 229
Vary: Accept-Encoding
Content-Type: text/html
```

Ahora comprobar la aplicación

```
david@Despliegue:~$ curl -I http://localhost/running-zones/04ArrayBootstrap.php
HTTP/1.1 200 OK
Date: Thu, 12 Nov 2026 17:58:09 GMT
Server: Apache/2.4.66 (Ubuntu)
Content-Type: text/html; charset=UTF-8

david@Despliegue:~$
```

Por último comprobar que PHP se ejecuta correctamente

```
david@Despliegue:~$ curl -s http://localhost/running-zones/04ArrayBootstrap.php | head -20
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Ejercicio 4</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha
384-sRIl4kxILFvY47J16cr9ZwB07vP4J8+LH7qKQnuqkuIAvNWLzeN8tE5YBujZqJLB" crossorigin="anonymous">

</head>
<body class="bg-secondary">

    <h3>Mostrar el primer día </h3>Lunes<h3>Mostrar consecutiva </h3>Lunes, Martes, Miércoles, Jueves, Viernes, Sábado,
Domingo,     <h3>Mostrar consecutiva en forma de lista</h3>
    <ol class="list-group list-group-numbered">
    <li class='list-group-item bg-primary text-white'>Lunes</li><li class='list-group-item bg-primary text-white'>Martes
</li><li class='list-group-item bg-primary text-white'>Miércoles</li><li class='list-group-item bg-primary text-white'>J
ueves</li><li class='list-group-item bg-primary text-white'>Viernes</li><li class='list-group-item bg-primary text-white
'>Sábado</li><li class='list-group-item bg-primary text-white'>Domingo</li></ol><br>    </ol>


</body>
</html>david@Despliegue:~$
```

Comprobación

